Hashan Fernando

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Professional Summary

Data Scientist and Researcher with 6+ years of experience applying Bayesian statistical modeling, machine learning, and NLP to healthcare and public health problems. Experienced in extracting, transforming, and curating real-world and clinical data into research-ready datasets and in building reproducible, well-documented analysis pipelines. Comfortable translating sophisticated technical research into practical, scalable solutions, with hands-on work spanning deep learning, transformer architectures, and end-to-end ML workflows.

Skills

Languages: Python, R, SQL, C, C++, Java

Statistics & Methods: Bayesian hierarchical & spatial modeling, MCMC, causal inference, machine learning, deep learning, NLP, LLMs, operations research, topological data analysis

Data & Databases: SQL & relational databases, data extraction/ETL, data cleaning & QA, REST/Web APIs, Spark

Dashboards & Visualization: Tableau, Looker (Data Studio), React

Frameworks & Tools: PyTorch, TensorFlow, scikit-learn, RStudio, Git, GCP, AWS

Education

Haslam College of Business, University of Tennessee, Knoxville, TN — 2022 – 2026
M.S. Statistics - (GPA: 4.00 out of 4.)

Postgraduate Institute of Science, University of Peradeniya, Sri Lanka — 2020 – 2022
M.S. Data Science - (GPA: 3.75 out of 4.)

Faculty of Science, University of Peradeniya, Sri Lanka — 2015 – 2018
B.Sc. Statistics and Operations Research

Experience

Data Science Researcher — Aug 2022 – Apr 2026
Advanced Computing in Health Sciences Section, **Oak Ridge National Laboratory**, Oak Ridge, TN

- Extracted, cleaned, and integrated VA electronic health record and mortality data, mapping ICD diagnosis and cause-of-death codes to standardized risk categories to build research-ready datasets for county-level opioid overdose modeling.
- Designed a novel algorithm integrating topological data analysis and Bayesian spatial modeling to identify complex spatial patterns in opioid overdose risk across the USA, reducing uncertainty in baseline Bayesian spatial model estimates.
- Built custom multi-channel CNN architectures in PyTorch to classify regional health risk strata across the USA using high-dimensional topological persistence images.

Researcher — Dec 2019 – Feb 2022
Department of Statistics and Computer Science, **University of Peradeniya**, Sri Lanka

- Developed a smart seal and audio detection system for protecting steel cargo from tampering, leveraging IoT sensors for real-time monitoring.

Marketing Analytics Consultant (Freelance) — June 2020 – Feb 2022

- Designed and analyzed marketing performance and sales dashboards (React, Google Data Studio) pulling multi-channel spend data via Supermetrics, Shopify, and the Facebook Ads API to identify trends and inform client optimization decisions.
- Managed client relationships and reporting cycles independently.

Data Scientist Intern — Jan 2019 – Jul 2019
VizuaMatix, Colombo, Sri Lanka

- Engineered a machine learning model to identify non-performing customers in a banking system.
- Developed a route optimization pipeline for food distribution trucks to improve efficiency and reduce operational costs.

Planning Intern — Oct 2018 – Jan 2019
MAS Holdings, Biyagama, Sri Lanka

- Developed multiple VBA applications for planning inventory.

Selected Projects & Competitions

- **1st Place Winner — MOSSAIC Hackathon**
  ORNL, National Cancer Institute & Leidos Biomedical
  Co-led a research team and empirically demonstrated data privacy leakage in cross-silo Federated Learning architectures by deploying gradient-based adversarial attacks against text transformer models.
- **Clinical Document Classification & NLP Clustering**
  COSC-526: Data Mining & Analytics Project
  Developed an end-to-end NLP pipeline on unstructured clinical notes; generated contextual embeddings for supervised classification and applied NER, Mapper-based TDA clustering, cosine similarity, and t-SNE/UMAP for medical terminology extraction and visualization.
- **Spatial - TDA**
  Developed a Python package designed for extracting topological information from spatial data.
- **Causal Inference for Marketing Analytics**
  BZAN-593: Independent Study
  Developed a Bayesian marketing mix model to estimate causal ROI of marketing spend from observational time-series data.

Publications

- **H. Fernando**, "Unmeasured Spatial Risk in Disease Modeling: A Topological Data Analysis Approach", *Manuscript in preparation*, 2026
- A. Deas, **H. Fernando**, A. Heidi, A. Kapadia, J. Trafton, A. Spannaus, V. Maroulas, "Identifying spatiotemporal patterns in opioid vulnerability: investigating the links between disability, prescription opioids and opioid-related mortality", *BMC Public Health 25*, no. 1 (2025): 1759 [pdf]
- **H. Fernando**, A. Spannaus, A. Kapadia, "A Geospatial Study of Opioid Overdose Risk in Diverse Regions", *Conference Proceedings for International Population Data Linkage Conference*, vol. 9, no. 5, 2024 [pdf]

Presentations

- Unmeasured Spatial Risk in Disease Modeling: A Topological Data Analysis Approach, UT-Oak Ridge Innovation Institute's Second Annual Research Symposium, Knoxville, TN, Feb 2025
- A Geospatial Study of Opioid Overdose Risk in Washington D.C. Metropolitan Area, Interdisciplinary Association for Population Health Science Conference, St. Louis, MO, Sep 2024
- A Geospatial Study of Opioid Overdose Risk in Diverse Regions, International Population Data Linkage Network Conference, Chicago, IL, Sep 2024

Awards

- Merit Award for M.S. in Data Science
- Half Colors, Colors Awarding Ceremony 2017, University of Peradeniya, Sri Lanka